Fiscal Year:	FY 2014	**Task Last Updated:**	FY 02/05/2014
PI Name:	Wiese, Claudia Ph.D.		
Project Title:	HZE-induced skin carcinoma: an in-vivo-like multicellular epithelial model system for analysis of radiation quality effects and DSB repair processes		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:	For information purposes only--PI moved in June 2014; however, funding did not go with her. As of 6/1/14, Colorado State University, Department of Environmental and Radiological Health Sciences, 485 MRB - 1618 Campus Delivery, Fort Collins, CO 80523-1618, Office: (970) 491 7618, Email: Claudia.Wiese@colostate.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
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Flight Program:			
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Task Description:	There is considerable evidence that ionizing radiation causes non-melanoma skin cancer (NMSC), and the risk for NMSC in astronauts is increased. To further assess the cancer risk for humans from space radiation, an in-vivo-like multicellular skin model system is used. Multiple different endpoints as surrogate markers for skin cancer are assessed. Persistent chromosomal damage, altered cell proliferation, and evasion from apoptosis are studied in 2D monolayer cultures of normal primary and of initiated human keratinocytes exposed to a variety of charged nuclei (Aim 1). In addition, to more realistically recapitulate the complex architecture of a tissue, an in-vivo-like skin model system, composed of fibroblasts and keratinocytes, is used to ascertain the risk of space radiation-induced cancer development processes in 3D (i.e. persistent chromosomal damage, altered cell proliferation, evasion from apoptosis, and differentiation) (Aim 2). Last, a surface transplantation assay of exposed 3D reconstructed human epithelia onto immunodeficient mice will be carried out in future to test for intermediate and late events in cancer progression (cell invasion and metastasis, respectively) (Aim 3). We hypothesize that, in basal cells, unfaithful DNA DSB repair will lead to persistent chromosomal damage, promoting altered cell growth and escape from tissue homeostasis. We believe that the proposed approach is highly relevant for assessing the risk of skin cancer in astronauts and for modeling space radiation-induced tumor development in other epithelial tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Skin as a model system to study radiation-induced human epithelial cancers: Skin is one of the most complex organs of the body, consisting of multiple cell types, including epithelial keratinocytes and connective tissue (i.e. fibroblasts). Keratinocytes form a stratified squamous epithelium and are composed of one basal cell layer and several differentiated cell layers. Proliferation is limited to the basal layer, whereas differentiating cells withdraw from the cell cycle and express various differentiation markers, dependent on the stage of their differentiation. Because radiation-induced neoplasms in skin derive primarily from epithelial cells, radiobiological studies on keratinocyte monolayers (2D), as suggested by us, provide a good first approximation for assessing cancer development processes triggered by ionizing radiation, but also by other environmental mutagens. Although 2D in vitro assays are high throughput and can be very informative, they do not appropriately recapitulate the complex interactions between different cell types and the extracellular matrix within tissues. The in vivo animal model systems, may provide the complex native environment, but are not well suited for experimental manipulation and may not accurately recapitulate the human condition. Three-dimensional (3D) in vitro organotypic culture models of epithelial tissues, like the one developed by us, provide an intermediate solution by recapitulating the major features of normal 3D tissue organization using human derived cells, and hold great potential for studying the effects of radiation and other environmental mutagens on tissue homeostasis. In this model system, keratinocytes are grown on a collagen layer containing fibroblasts and are exposed to air. These conditions allow fibroblasts and keratinocytes to interact such that a stratified and differentiated epithelium is formed, resembling the architecture of the human skin. For the experiments proposed in our study, we use low-passage normal primary human keratinocytes (NPKs). The advantage of primary keratinocytes is that their response most closely resembles that of normal skin. To mimic intraepithelial neoplastic lesions, we also assess immortalized non-tumorigenic HaCaT cells. HaCaT cells were derived from normal skin of a male patient and spontaneously immortalized in culture. However, HaCaT cells retain the functional differentiation properties of normal keratinocytes, form a stratified squamous epithelium when grown in organotypic cultures (despite the absence of fully functional p53), and are non-tumorigenic when injected in mice. Notably, HaCaT cells carry a UVB fingerprint mutation in p53 that resides in many skin cells of healthy middle-aged individuals. This model for intraepithelial neoplasia is highly relevant since it has been shown previously that adult skin contains clonal patches of p53 mutated keratinocytes approximately 60-3000 cells in size, which can involve up to 4% of the epidermis. We believe that our studies are relevant for a better understanding of the processes involved in the development of epithelial cancers due to exposure to environmental mutagens. The skin model system consisting of co-cultured fibroblasts and keratinocytes is one of the best-understood human tissue systems, and fibroblast-derived soluble factors play an important role in maintaining the well-controlled balance between epithelial proliferation and differentiation. For human skin, many of these paracrine factors and the pathways they are involved in are known, and they also are conserved across other epithelial human organs. Therefore, the response of skin tissue to environmental factors is relevant for predicting cancer risk in other tissues of epithelial origin.
Task Progress:	Our study is aimed at further elucidating the Relative Biological Effect (RBE) of charged particle radiation, encompassing the Linear Energy Transfer (LET) range from 44-250 keV/µm. To assess genotoxicity from low doses of different radiation qualities the cytokinesis-block micronucleus assay is used. This assay is high-throughput and highly sensitive, and allows the accumulation of all cells at the bi-nucleate stage, regardless of their division kinetics. Importantly, in humans, micronuclei (MN) are one of the four main endpoints, together with chromosomal aberrations, aneuploidy, and sister chromatid exchanges (SCE), in the identification of cancer initiation processes [1, 2]. Normal primary human keratinocytes and immortalized HaCaT keratinocytes, grown in 2D cell monolayers, were exposed to graded doses of 300 or 1000 MeV/n Si or Fe ions and MN frequency was assessed. The advantage of using primary keratinocytes is that their response to IR most closely resembles that of normal human skin cells. In addition, immortalized HaCaT cells were used as a model for intraepithelial neoplasia. HaCaT cells carry a UVB fingerprint mutation in their p53 gene [3], also observed in sun-exposed areas of the skin of healthy middle-aged individuals, such as astronauts. Although 2D in vitro assays are high-throughput and can be very informative, they do not appropriately recapitulate the complex interactions between different cell types and the extracellular matrix within tissues. For this reason, to more realistically simulate the complex architecture of the skin, we have also investigated 3D reconstructed multi-cellular organotypic skin epithelia, in which the keratinocytes form a stratified squamous epithelium and a proliferative basal cell layer when cultured on top of a fibroblast contracted collagen matrix. These tissues were exposed to the same radiation qualities as described above, and MN frequency was assessed in the proliferative keratinocyte compartment. We have established dose response curves for MN induction of 300 and 1000 keV/µm Si and Fe ions. 137Cs gamma-rays were used as low LET reference radiation. Although the shapes and magnitudes of the dose response curves differ, exposure to 300 and 1000 MeV/n Fe ions led to the highest fraction of MN induced in both primary and initiated keratinocytes grown in 2D. RBEmax values obtained range from 2.8 to 5.6 and from 2.9 to 7.2 for primary and initiated keratinocytes, respectively. At radiation doses = 0.4 Gy, 300 MeV/n Fe ions were most effective in inducing MN in normal keratinocytes. However, at doses larger than 0.5 Gy, for 300 MeV/n Fe ions an “over-kill effect” was observed, and 1000 MeV/n Fe ions were more efficient in inducing MN. In initiated keratinocytes, 300 MeV/n Fe ions produced MN with greater frequency than 1000 MeV/n Fe ions up to 0.8 Gy. Compared to primary keratinocytes, MN formation was higher in immortalized keratinocytes for all doses and radiation qualities investigated, reflecting the

	greater tolerance of persistent DNA damage in initiated cells. Compared to keratinocytes grown in 2D, RBEmax values for MN induction were generally lower in keratinocytes recovered from irradiated 3D skin epithelia and, for Fe ion exposure, ranged from 1.5 to 1.6 and from 1.6 to 4.0 for primary and initiated epidermal keratinocytes, respectively. Our results suggest that tissue architecture plays an important role in preventing the persistence of genotoxic lesions from space radiation exposure, possibly by enhancing DNA repair fidelity. 1. Hagmar L., et al. 2001, Int. J. Hyg. Environ. Health 204: 43-7. 2. Tucker J.D. and Preston R.J. 1996, Mutat. Res. 365: 147-59. 3. Lehman T. A., et al. 1993, Carcinogenesis 14:833-9.
Bibliography Type:	Description: (Last Updated: 04/11/2018)
Abstracts for Journals and Proceedings	Snijders AM, Mannion BJ, Leung SG, Rydberg B, Kronenberg A, Wiese C. "An In-Vivo-Like Multi-Cellular Skin Model System for the Analysis of Radiation Quality Effects." Presented at the HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. Poster Presentation; Program and Abstract Book. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Abstracts for Journals and Proceedings	Snijders AM, Mannion BJ, Leung SG, Moon S, Kronenberg A, Wiese C. "Micronucleus formation in human skin keratinocytes exposed to different radiation qualities in 2D and in 3D." Oral presentation in session entitled: Cancer Risks II; 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. Program Book; 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3312.pdf , Feb-2014